Inequalities — problem 7.98

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Let $n > 2$ and $x_1, \ldots, x_n$ be nonnegative reals. Prove the inequality

$$(x_1 x_2 \cdots x_n)^{1/n} + \frac{1}{n} \sum_{i<j} |x_i - x_j| \geq \frac{x_1 + \cdots + x_n}{n}.$$